Safe-Seat Washington

Two questions, measured separately: was the general election close, and did it offer a choice between the parties? (Observed, 2016–2024)

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AI-assisted drafting and analysis review. All figures are reproducible from public-record data and from the open-source scripts cited below, including `scripts/diag_seat_competition.py`, which builds the seat universe from certified statewide returns and fails loudly if any cycle does not reconcile — the House and U.S. House against their statutory sizes, the Senate seat by seat against the roster its staggered terms imply. The paper source, code, and data-acquisition recipe are public at https://github.com/skirby359/who-decides. Contact: kirby@tikorconsulting.com.

DRAFT — pending human/editorial sign-off. *`scripts/verify_safe_seat.py` scrapes this paper and asserts its figures against the data, with the exceptions the script names in its own output; that gate is automated and is not the sign-off. The sign-off is a person reading the paper end to end, recorded in `safe-seat-submission-notes.md` §Sign-off.*

Companion to "Who Decides Washington State?" and the electoral-health white paper (Finding 2). Where the lead paper showed who *turns out, this one asks what their ballot actually offered.*

> **Revision note (2026-07-27).** This paper was substantially rebuilt after an adversarial review. Three defects were confirmed and corrected: the seat universe was incomplete for 2016 and 2018 (King County was absent from the statewide precinct files, costing 24 House seats per year); the "same-party" category was misclassified (the rule captured *any* race lacking a D-vs-R pairing, including D-vs-independent); and two distinct questions — whether a race was close, and whether it offered a partisan choice — were collapsed into one number. Headline figures have changed and Appendix G records the before/after. The claim that seats were "decided before November" has been withdrawn as unsupported by this design.
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> **Revision note (2026-07-28).** A second internal pass, this one auditing the paper against its own scripts rather than its argument, corrected five figures and closed one reproducibility gap. The 2020 partisan-availability share moves 27.6% → **26.9%** (a misspelled party string in the certified file had made a real Democrat-versus-Republican race read as no-choice); the 2020 primary/general median moves 61.6% → **61.5%**; New York's ≥12-point cell moves 85.2% → **85.9%**; and Appendix E's exploratory seat/vote gaps move to **WA +2.1 / TX +3.3**, the Texas figure having been computed on the same imputed party split Appendix F retires. The five-cycle primary/general table now has a derivation in `diag_seat_competition.py` instead of resting on an unrecorded computation. No headline figure changed. Appendix G records the detail.

Abstract

Claims that most legislative seats are "safe" usually rest on a projected margin, which invites the rebuttal that the model is wrong. This paper discards the projection and counts observed general-election results for every partisan legislative and congressional seat on Washington's ballot from 2016 through 2024, then repeats the count in three comparison states. It measures two things separately, because they are different questions. **Candidate competition** asks whether the race was close, using the margin between the top two candidates regardless of party. **Partisan availability** asks whether the ballot offered both a Democrat and a Republican. Washington's top-two primary makes the distinction unavoidable: a general election can be a genuine contest between two Democrats, and it can also present a single unopposed candidate. In 2024, 111 of 133 seats (83.5%) were not close — decided by ten points or more, or uncontested — and 46 (34.6%) offered no Democratic-versus-Republican option. The two overlap but are not the same: of sixteen same-party generals, fifteen were also lopsided, but one was decided by six points. Across five cycles the not-close share runs 79–88% and the no-major-choice share 25–49%. The pattern is not confined to Washington: in the lower chambers of three comparison states, 88–94% of seats were not close. Safe seats are bipartisan, splitting 68 Democratic to 43 Republican in Washington in 2024. The conclusion does not turn on the competitiveness threshold: the not-close share stays between 74% and 98% across cuts from 15 points to 5, so a large majority of seats are not close at every setting tested. The paper counts contests and margins; it does not establish when the binding choice occurred, and makes no claim about which party benefits.

Keywords: electoral competition; safe seats; uncontested elections; top-two primary; state legislatures; primary elections; Washington; New York; Texas; Idaho.

The question, and why "observed" matters

The standard "most seats are safe" claim usually rests on a *projection* — a model's predicted margin. That invites the obvious rebuttal: *your model could be wrong.* So this paper throws the projection out and counts the **actual** general-election result of every partisan legislative and congressional seat on Washington's ballot, 2016–2024. The unit is the **seat** — each State Representative position, each State Senator race up that cycle, each U.S. House race — the thing a voter actually marks.

Washington's **top-two primary** forces a distinction that most safe-seat work can elide. Because the two highest primary finishers advance regardless of party, a general election can be:

- **a single candidate**, with no opponent at all;
- **two candidates of the same party**, offering a real choice between people but none between parties;
- **a Democrat against a Republican**, the conventional case;
- **a major-party candidate against a minor-party or independent one.**

A 51–49 general between two Democrats is a contest. It is also a ballot with no cross-party option. Collapsing those facts into a single "non-competitive" number answers neither question well, so this paper reports two dimensions throughout:

dimension	question	measure
Candidate competition	Was it close?	margin between the top two candidates, any party
Partisan availability	Was there a D-vs-R choice?	which parties appeared on the ballot

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The seat universe

Because an earlier version of this analysis under-counted, the universe is now built from the **certified statewide summary returns** and reconciled before anything is computed: the House and U.S. House against fixed statutory sizes, the Senate against the roster its four-year staggered terms imply. Washington elects all 98 House positions and 10 U.S. House members every even year; the Senate elects 25 districts in presidential years and 24 in midterms, plus any specials.

year	House (want 98)	Senate	U.S. House (want 10)	total
2016	98	26	10	134
2018	98	25	10	133
2020	98	26	10	134
2022	98	25	10	133
2024	98	25	10	133

`scripts/diag_seat_competition.py` exits non-zero if any cycle fails this reconciliation. The prior approach derived races from the precinct-results table, where a race absent from the data simply disappeared rather than being detected as missing; Appendix G documents what that cost.

What is asserted, and how. The House and U.S. House are checked against **fixed statutory expectations** — 98 and 10 — that do not come from the file being checked, so a missing race fails the run. The **Senate is now checked by district identity**, not by count.

Washington's 49 Senate districts run four-year staggered terms, which partitions them into two alternating cohorts: **25 districts elect in presidential years, 24 in midterms**, and every district appears exactly once every four years. That periodicity is a property of the term system rather than of any particular file, so the roster expected in each cycle can be written down in advance — it is, in `docs/reference/wa_senate_cycle_2016-2024.csv` — and the certified file checked against it seat by seat. Four races fall outside the regular rotation (LD 36 in 2016, LD 39 in 2018, LD 38 in 2020, LD 27 in 2022); each is listed explicitly.

Identity matters here rather than count, because a count check cannot see a substitution. All five cycles reconcile exactly, and the check is verified to catch both a dropped district and a count-preserving swap.

Two honest limits on that claim. First, the cohort assignment was **derived from these same five certified files** and then cross-checked against the term structure — 25 + 24 = 49, disjoint, every district once per four years, every cycle's roster reproduced exactly. It was not taken from an external roster of seats up for election. So this closes the loop for *future* files, which is what a gate is for; it is not independent confirmation that these five files were themselves complete. Second, the

four off-cycle races are **consistent with** special elections to fill vacancies, but the certified files do not say so — the race name is just "State Senator" — so that reading is an inference from periodicity rather than something the source states.

Dimension 1 — was the race close?

Margin between the top two candidates, regardless of party. "Not close" means a single candidate, or a margin of ten points or more.

year	seats	single candidate	Tossup <5	Lean 5–10	Likely 10–20	Solid 20+	not close
2016	134	27	8	8	32	59	**88.1%**
2018	133	15	19	9	24	66	**78.9%**
2020	134	14	11	11	28	70	**83.6%**
2022	133	24	10	8	31	60	**86.5%**
2024	133	23	10	12	23	65	**83.5%**

- **Defensible claim.** In a typical Washington cycle **roughly five in six legislative and congressional seats are not close**. In 2024, of 133 partisan seats only **22 (16.5%) were decided by under ten points**. The share runs 79–88% across a decade, with **2018 the least lopsided year** — the blue-wave cycle, when 19 seats landed inside five points, the most in the series and **nearly double the next highest, 11 in 2020**.
- **Safe seats are bipartisan.** Among 2024's not-close seats, **68 were won by Democrats and 43 by Republicans** — consistent with a geographically sorted electorate, a mechanism this design does not estimate. (Winner party is taken from the leading candidate, not inferred from aggregate party vote totals; Appendix G explains why that distinction mattered.)

Dimension 2 — did the ballot offer a partisan choice?

year	seats	D-v-R	D-v-D	R-v-R	D-v-other	R-v-other	single	no D-v-R
2016	134	69	4	5	15	14	27	**48.5%**
2018	133	98	6	0	13	1	15	**26.3%**
2020	134	100	8	1	7	4	14	**25.4%**
2022	133	86	10	6	2	5	24	**35.3%**
2024	133	87	8	8	5	2	23	**34.6%**

- **Defensible claim.** In 2024, **46 of 133 Washington races — more than a third of the partisan ballot — offered no Democratic-versus-Republican option**: 23 had a single candidate, 16 pitted two candidates of the same party against each other, and 7 set a major-party candidate against a minor-party or independent one. This count is **threshold-free**: no margin cutoff enters it.
- **The single-candidate count is the hardest number in the paper.** Twenty-three seats in 2024 presented voters with exactly one name. That is not a judgment about competitiveness; it is a headcount. Both implementations exclude candidates with zero recorded votes, so "one candidate" and "one name on the ballot" are the same proposition only if no ballot-listed, non-write-in candidate drew zero votes. **There are none, in any of the five cycles** — the verifier measures it and prints `zero-vote ballot candidates = 0`, and fails if that ever stops being true.
- 2016 is the outlier at 48.5%, and partly for a definitional reason — **eight** distinct party strings appeared on that year's ballot that name or gesture at a major party without qualifying as one under the published rule: six Independent-flavoured ("Indep't Democrat", "Independent Dem", "Independent Dem.", "Independent GOP", "Independent", "Independent Rep") plus **two** hybrids naming two parties at once ("GOP/Independent" and "Dem/Working Fmly"). These are the strings the classification rule has to adjudicate; 2016 also carried ordinary minor-party and no-preference labels ("Libertarian", "States No Party Preference", "Non-Partisan"), which raise no such question and are not counted

here. Earlier versions of this sentence said *seven*, listed only the first hybrid, and described the eight as though they were every non-major string on the ballot. See the sensitivity test below.

The two dimensions are not the same question

Cross-tabulating them for 2024 shows where they agree and where they part:

	single	Tossup <5	Lean 5–10	Likely 10–20	Solid 20+
D-v-R	0	10	11	20	46
D-v-D	0	0	0	1	7
R-v-R	0	0	1	2	5
D-v-other	0	0	0	0	5
R-v-other	0	0	0	0	2
single candidate	23	0	0	0	0

Most same-party generals are also lopsided — 15 of 16 in 2024 exceeded ten points, and 12 exceeded twenty. But **one was a genuine contest**: Washington's 4th Congressional District, an R-vs-R general decided by **6.0 points** (Dan Newhouse). Treating that race as "non-competitive" because it lacked a Democrat would be wrong. It is a competitive election without a partisan choice, and the two-dimension design is what makes it visible.

Sensitivity: how party preference is read

Washington's top-two system has no nominees, only stated preferences, and the preference is **informational**: a candidate may write essentially any string, and it implies no nomination, endorsement or association by the party named. That matters for how the results below should be read. When this paper counts a Republican-versus-"Culture Republican" general as a same-party race, it is not asserting that the state treats those as one party. It is applying a **researcher-defined party-family category**, and the honest response to that is to report the alternatives rather than defend the choice.

Three specifications are therefore enumerated string by string in `docs/reference/wa_party_strings_2016-2024.csv`, each a superset of the one before:

- **Literal** — orthography only. `G.O.P.`, `G.O.P` and `R` normalise to Republican because they spell the party's own name; faction names stay distinct, so "MAGA Republican" and "Culture Republican" are *other*.
- **Family** — **the published rule.** Faction qualifiers fold into the major party; independence qualifiers and two-party hybrids do not.
- **Expansive** — also folds "Independent Dem.", "Ind. Republican", "GOP/Independent" and "Dem/Working Fmly" into their major party.

year	literal	family (published)	expansive
2016	48.5%	**48.5%**	38.1%
2018	26.3%	**26.3%**	24.8%
2020	25.4%	**25.4%**	23.9%
2022	35.3%	**35.3%**	35.3%
2024	35.3%	**34.6%**	34.6%

The conclusion does not turn on the choice. Literal and family agree in four of the five cycles and differ by three-quarters of a point in the fifth — 2024, where exactly one faction-qualified string ("MAGA Republican", in Congressional District 2) changes a race from major-versus-other to D-v-R. The expansive column moves 2016 by ten points and nothing else by more than 1.5, because 2016 is the year the independence-flavoured labels cluster in.

One historical note, because it explains the shape of the correction. The regex this enumeration replaced was **not** any of these three: it treated faction names as other, like *literal*, but missed the orthographic variants that *literal* normalises. That is why the 2024 literal column reproduces the pre-audit published figure of 35.3% while 2018 and 2020 do not.

The rule matters in 2016 and is close to immaterial elsewhere. Both specifications are now **enumerated string by string** in `docs/reference/wa_party_strings_2016-2024.csv` rather than expressed as regexes, and the only judgment the test still varies is whether independence-qualified strings and two-party hybrids fold into a major party. That is why the 2024 delta is 0.0: the faction-qualified strings that used to separate the two specifications ("MAGA Republican", "Culture Republican") are major under both. Deltas are computed on unrounded shares, so they need not equal the difference of the two printed columns (2016: 48.5075 − 38.0597 = 10.4478).

Spelling and abbreviation are handled as data, not judgment. The 2020 certified file records the Democratic candidate in Legislative District 8, Position 1 as preferring the "Democractic" party — a misspelling, not a distinct preference. Read literally it made a genuine Democrat-versus-Republican general (Klippert 51,981, Regev 26,979) classify as Republican-versus-other, and correcting it alone moved the 2020 no-D-v-R share from 27.6% to 26.9%. Such corrections apply in **both** the strict and the loose specification, because the sensitivity test is only informative if it varies the interpretive rule while holding the transcription constant. Hybrids like 2016's "GOP/Independent" are real stated preferences and are left in the *other* bucket, where the loose column prices them.

That single correction is no longer the whole story, and the reason is worth stating. A 2026-08-08 audit enumerated every distinct party string in all five certified files — 32 of them — instead of trusting a regex to match the ones anybody had thought of. It found five further strings naming a major party that were being counted as minor: `G.O.P.` (2018), `G.O.P` without the trailing period (2016), `R` twice (2020), `MAGA Republican` (2024) and `Culture Republican` (2024). Six races were misclassified across four cycles. Correcting them moves 2018 to **26.3%**, 2020 to **25.4%** and 2024 to **34.6%**, and widens the five-cycle range to 25–49%. **Dimension 1 is untouched** — margins do not depend on party — so the headline 111 of 133 stands unchanged.

The defect was never any one pattern; it was that a regex cannot report what it failed to match. Classification is therefore no longer a regex at all. Both specifications are enumerated string by string in `docs/reference/wa_party_strings_2016-2024.csv`, each row carrying its reason, and a string absent from that file is a hard error rather than a silent "other". The rule the enumeration encodes: a string names a major party iff it carries a variant of that party's own name and is not qualified by independence. Faction qualifiers do not disqualify; independence qualifiers and two-party hybrids do. The independence exclusion is what keeps the classification symmetric — a rule reading any string containing "Republican" as Republican, while leaving "Independent Dem." as other, would flip nine Republican-flavoured races and no Democratic ones.

The four-state comparison

The same count against each state's **lower chamber** — the one body fully up every cycle everywhere. The comparison is **unit- and definition-aligned but not fully time-aligned**: chamber, unit, margin definition and threshold are identical, while New York's most recent loaded Assembly general is **2022** against 2024 for the other three. Completeness is reported explicitly rather than assumed. *(This paragraph said "like-for-like" until 2026-08-16, which overstated an alignment the next bullet already qualified.)*

state (chamber)	loaded / expected	not close	no D-v-R
WA House 2024	98 / 98	**87.8%**	39.8% (39)
NY Assembly 2022	149 loaded + 1 certified supplement (AD-23)	**88.0%**	32.0% (48)
TX House 2024	96 canvass + 54 certified single-candidate	**94.0%**	40.7% (61)
ID House 2024	70 / 70	**92.9%**	28.6% (20)

- **Defensible claim.** A low not-close share is **not a Washington peculiarity** — in every state examined, **88–94% of lower-chamber seats were not close**, blue and red alike, and **more than a quarter offered no D-vs-R option.**
- **New York is now complete, and the one absent seat turned out to be the chamber's closest.** The Assembly has 150 districts and the loaded returns carry 149; a check confirms the missing one is exactly Assembly District 23 and nothing else. Earlier drafts bounded the effect rather than resolving it, reporting 149 of 150 seats and a range wide enough to cover either outcome for the absent one. That was unnecessary: the seat is not unknowable. The New York State Board of Elections publishes the certified contest, and AD-23 was decided by **fifteen votes** — Stacey G. Pheffer Amato **16,185** to Thomas P. Sullivan **16,170**, a margin of 0.046 points and the narrowest race in the chamber. Supplying it makes the denominator the real 150, and the chamber reads **88.0%** not close with **32.0%** offering no D-vs-R option. Figures and provenance are pinned in `docs/reference/ny_ad23_2022.csv`; the totals are candidate totals across all ballot lines, which is not the same as the Democratic and Republican lines alone.

- **Texas is 64% loaded in the canvass returns**, and its figures depend on a backfill of the 54 uncontested seats that source omits. That backfill is now **verified seat by seat against the Secretary of State's certified results**, which independently show exactly those 54 districts as single-candidate, and supply the observed winning party for each (Appendix F). Its not-close seats split **56 D / 85 R**.
- **The comparison states are scored by a shortcut, and it was checked rather than assumed.** Washington's "not close" comes from the top-two margin, so a *contested* same-party general under ten points counts as close — the WA-04 case above. The comparison-state code takes any seat with no D-vs-R option as not close whatever its margin, which would inflate their figures if such a seat were ever competitive. Re-scoring every no-major-choice seat in the three states on its top-two margin finds **none under ten points** — 0 of 48 in New York, 0 of 61 in Texas, 0 of 20 in Idaho — so the three percentages are identical under either rule, so the coding choice is aligned in fact and not merely in intent. This is a measured result, not a property of the definitions, and `diag_safe_seat_robustness.py` re-measures it on every run.
- Most recent loaded general: WA/TX/ID 2024, NY 2022.

Why it matters

The lead paper showed the off-year electorate is half-sized and older than the presidential one. This paper shows that even in the high-turnout even-year general, most legislative and congressional seats are not close, and a third of the ballot offers no choice between the parties.

Where a general election is not close, attention naturally turns to the August top-two primary — the obvious place to look for whether meaningful competition occurred earlier, though this design cannot establish that it did. **The primary is much the smaller round.** Matching every seat to its own August primary on the corrected universe, the median primary race drew this share of the votes cast in the same seat's general:

2016	2018	2020	2022	2024
42.1%	55.9%	61.5%	61.2%	**51.2%**

Read this as a ratio of **votes cast in a contest**, not of distinct voters: roll-off and undervoting mean a race's vote total is not a headcount of participants. So in 2024 the median legislative or congressional primary recorded about half as many votes as the November contest for the same seat.

That the primary is smaller does not establish that it is where the decision was made. Observed November margins can show a race was not close; they cannot show *when* the binding choice occurred, or that a primary presented a meaningful alternative. Establishing that would require tracing each seat through its primary — whether it was contested, whether the eventual winner faced a credible same-party rival, and whether the general's finalists were themselves closely matched. That is the obvious sequel and is not attempted here.

What the evidence does support is narrower and still substantial: **a large share of Washington's legislative general elections ended with margins of ten points or more, and for roughly a third of seats the ballot offered no choice between the two major parties.** The earlier wording here said those seats were "settled in November by margins wide enough that the result was not in doubt" — which smuggles an *ex ante* claim about expectations back into an *ex post* measure. A race that was genuinely uncertain in September can finish 60–40; the margin records how it ended, not what was in doubt. Whether alternative primary or general-election structures — top-four, ranked choice — would increase candidate competition or cross-party choice is a question these findings motivate, not one they answer.

What this paper does not claim, and limits

- **It does not establish that seats were "decided before November."** That inference appeared in an earlier version and has been withdrawn. Observed general-election margins cannot date the binding decision.
- **"Not close" is not "illegitimate."** A forty-point margin can mean the voters there genuinely agree. The finding is strongest for the 23 single-candidate seats and the 16 same-party generals, and weakest as a democratic-deficit reading for lopsided D-vs-R contests. Appendix A takes this at full strength.
- **The partisan analysis is descriptive.** Safe-seat party totals and the seat/vote comparison in Appendix E describe patterns; they do not establish causation, intent, or gerrymandering. An earlier version claimed to make "no partisan-

consequence claim" while analysing exactly that — the contradiction is resolved by scoping the claim rather than denying the analysis.

- **The Texas backfill is verified and its party split is now observed.** Earlier versions imputed holding party from presidential lean, which proved wrong in 5 of 54 seats. Party now comes from certified returns, and the split is 56 D / 85 R rather than the previously reported 51 D / 90 R. Appendix F documents the correction.
- **New York is a cycle behind** (2022). It is no longer missing Assembly District 23: that seat is supplied from the certified NYSBOE contest, so the chamber is complete at 150 and the former 88.0–88.7% bound is retired.
- **Margins are between candidates, not parties**, in Dimension 1. Third-party votes count toward the margin when a minor-party candidate is one of the top two, and are excluded from the denominator otherwise — which can only arise in the comparison states, since a Washington general carries at most two named, non-write-in candidates.
- **Primary participation is measured in race votes, not voters** — see Appendix C.

Appendices

Appendix A — The objections, in full

1. Not-close seats still represent their voters. A Solid-D Seattle seat and a Solid-R rural seat each *represent* the electorate that produced them; a wide margin can be agreement rather than foreclosure. This is the strongest objection and it is substantially correct for lopsided D-vs-R contests. It has far less force against the 23 seats that offered one name: no degree of voter agreement explains the absence of an alternative to agree or disagree with. The two-dimension design exists so that these cases are counted separately instead of averaged together.

2. The ten-point threshold is a knob. True, and Appendix E tests the whole plausible range: the not-close share moves between 74% and 98% across cuts from 15 points to 5, and never approaches "competitive" at any setting. Dimension 2 is threshold-free entirely.

3. Same-party generals are a Washington artifact of top-two. In states with party primaries the analogue is a seat with no major-party filer, and the four-state table folds both into one comparable bucket. But the deeper version of this objection — that a same-party general may be a real contest — is correct, was mishandled in the earlier version, and is now measured directly. One of Washington's sixteen same-party generals in 2024 was decided by six points.

4. The safe-seat split proves a gerrymander. It does not, and the paper declines the claim. Washington's not-close seats split 68 D to 43 R — 61.3% Democratic — against a 59.5% Democratic share of the two-party presidential vote, a gap of 1.8 points, close to proportionate. Chen & Rodden (2013) is the standing demonstration that residential geography produces seat/vote bias with nobody drawing it, and single-member districts have no proportionality expectation to begin with. Appendix E reports the comparison descriptively and draws no inference about intent.

5. The Texas number rests on rows you added. It did, and the objection prompted a proper check. The backfilled 54 have since been verified one by one against the Secretary of State's certified results, which independently identify exactly those districts as single-candidate and supply each winner's party. The headline was unchanged by the check (94.0% either way); the party split was not, and has been corrected. Appendix F sets out both. Excluding Texas entirely would still not change the paper's conclusion; it would remove the most extreme case.

6. Your earlier numbers were wrong. Also correct. Appendix G documents what changed and why, rather than quietly restating.

Appendix B — Data access and provenance

- **What the figures are computed from.** Certified statewide general-election summary returns, one row per candidate per race, carrying race name, candidate, party preference and vote total. These are **published aggregate election results** — this paper touches no voter file and no personal data, and carries no privacy surface.
- **Washington.** Secretary of State certified statewide summaries, files `20161108_AllState.csv`, `20181106_AllState.csv`, `20201103_AllState.csv`, `20221108_AllState.csv`, `20241105_AllState.csv`. These are the authoritative source for the seat universe. The corresponding `_AllStatePrecincts.csv` files are **not** used for the universe, because King County is largely absent from them in 2016 and 2018 — the defect Appendix G describes.
- **New York.** State Board of Elections certified returns, ingested via the OpenElections-NY compilation of the county canvasses — the intermediary an earlier version of this note said should be cited before publication; the citations file

below names the exact artifact, its checksum, and the loader commit.

- **Idaho.** Secretary of State certified returns.
- **Texas.** Two sources, because one is insufficient. The Legislative Council's canvass-grade VTD returns supply precinct-level results but **omit uncontested races at every stage**, carrying 96 of 150 House districts. The **Texas Secretary of State's election-night results service** (`results.texas-election.com` , read 2026-07-27) publishes all 150 including uncontested ones with candidate, party and votes; it is the seat universe and the source of holding party, captured to `data/raw/tx/2024_tx_house_candidates.csv` . The TLC r206 report (`planh2316_r206_election24g.xls`) is retained only for district presidential lean in Appendix E.
- **Dataset citations — assembled 2026-08-29.** `docs/reference/safe_seat_source_citations.csv` carries one row per source: originating authority, source URL, the exact ingested artifact, its SHA-256, the on-disk archive date, and the loader with its last commit. It also resolves the New York provenance note above: the ingested artifact is the OpenElections-NY compilation of the NYSBOE certified county canvasses (`20221108__ny__general__precinct.csv`), loaded by the generic results loader, with AD-23 supplied from the certified NYSBOE contest pinned in-repo. *(This bullet said the citations were "not yet assembled" until 2026-08-29; an external series review listed the gap among the paper's remaining pre-publication items.)*

Appendix C — Methods

- **Universe.** Built from certified statewide summary returns and reconciled before use: 98 WA House positions and 10 U.S. House against fixed statutory sizes, and the Senate **by district identity** against the roster implied by its four-year staggered terms (25 presidential-year districts, 24 midterm, plus four listed off-cycle races), pinned in `docs/reference/wa_senate_cycle_2016-2024.csv` . Identity rather than count, because a count cannot detect a substitution. `scripts/diag_seat_competition.py` exits non-zero on any mismatch. Write-ins are excluded; candidates with zero votes are excluded.
- **Party.** Taken from the certified "Prefers ___ Party" string, and resolved against a frozen enumeration of all 32 strings in the five certified files (`docs/reference/wa_party_strings_2016-2024.csv`); an unlisted string is a hard error, not a silent "other". A string counts as major when it carries a variant of the party's own name and is **not** qualified by independence. Faction qualifiers therefore do **not** disqualify — "MAGA Republican" and "Culture Republican" are major — while independence-qualified strings ("Independent Dem.", "Ind. Republican") and two-party hybrids ("GOP/Independent", "Dem/Working Fmly") are other. Spelling and abbreviation are normalized before classification in every specification: "Democractic" (2020), "G.O.P." (2018), "G.O.P" (2016) and "R" (2020). Three specifications are reported in the body's sensitivity test. **Because top-two has no nominees, this is a statement about stated preference, not nomination — and the party-family grouping is the researcher's, not the state's.**
- **"Safe seat"** is used throughout as shorthand for *a seat that was not close in the observed result*. It carries no forward-looking claim; nothing here projects a future margin, which is the point of the design.
- **Dimension 1, candidate competition.** Margin between the top two candidates by votes, regardless of party: |first − second| / (first + second). Bands: Tossup <5, Lean 5–10, Likely 10–20, Solid 20+. "Not close" = single candidate or margin ≥10. In Washington the general carries at most two named, non-write-in candidates — 23 seats in 2024 carried one — so this is the whole field; in the comparison states, where three or more can appear, votes for candidates below second place are outside the denominator and margins are correspondingly slightly wider.
- **Dimension 2, partisan availability.** Classified from the set of parties actually on the ballot: D-v-R, D-v-D, R-v-R, D-v-other, R-v-other, other-only, single candidate.
- **Winner party** is the party of the leading candidate. An earlier version inferred it by comparing aggregate D and R vote totals, which mislabels a race where both are zero; Appendix G records the fix.
- **Primary participation.** Each general seat is matched to its same-office, same-district August primary, and the reported figure is the **median ratio of primary race votes to general race votes** — a comparison of votes cast in a contest, *not* of distinct voters. Roll-off and undervoting affect race totals, so the two are not interchangeable. Recomputed on the corrected universe by `scripts/diag_seat_competition.py` , which reports the matched-seat count alongside each median and fails the run if any seat is unmatched (all five cycles match completely): 42.1 / 55.9 / 61.5 / 61.2 / 51.2% for 2016–2024. The 2024 and 2022 figures are within a point of the previously published ~51% and ~62%, so the correction to the seat universe did not move this measure.
- **Reproduction.** `scripts/diag_seat_competition.py` builds the classification and writes `reports/seat_competition.csv` (one row per seat, both dimensions). Supporting scripts: `diag_safe_seat_states.py` (four-state), `diag_tx_safe_seat_backfill.py` (Appendix F),

`diag_safe_seat_robustness.py` (Appendix E), `diag_safe_seat_party_ratio.py` and `diag_efficiency_gap.py` (Appendix E's exploratory cuts).

- **Where the scripts disagree with each other, and why.** Two of the supporting scripts predate the universe rebuild and can still reproduce their old Washington cell: `diag_safe_seat_states.py` reads the precinct table under the retired conflated definition and computes WA House at 88.8%, and `diag_safe_seat_robustness.py`'s threshold sweep reads the same table. Every Washington figure in this paper comes from `diag_seat_competition.py` on the certified universe (87.8%), which is authoritative. **Since 2026-08-29 both scripts withhold their superseded cells unless run with `--legacy`**, which prints them labelled — default output shows a withheld-cell marker beside the authoritative value, so the earlier published numbers remain reproducible for audit without sitting in default output where a casual read or a machine scrape would take them as current. (They printed by default, labelled, until an external series review pointed out that casually machine-readable obsolete output is how propagation errors start.) `diag_safe_seat_party_ratio.py` was corrected in place, since the paper quotes it directly.

Appendix D — Related work

- **The primary as the real election under one-party dominance.** V. O. Key Jr., with the assistance of Alexander Heard, *Southern Politics in State and Nation* (New York: Alfred A. Knopf, 1949). Hirano & Snyder, *Primary Elections in the United States* (Cambridge University Press, 2019), doi:10.1017/9781139946537.
- **The decline of competition in general elections.** Abramowitz, Alexander & Gunning, "Incumbency, Redistricting, and the Decline of Competition in U.S. House Elections," *Journal of Politics* 68(1) (2006): 75–88, doi:10.1111/j.1468-2508.2006.00371.x.
- **Uncontested seats as their own phenomenon.** Squire, "Uncontested Seats in State Legislative Elections," *Legislative Studies Quarterly* 25(1) (2000): 131–146. Burden & Snyder, "Explaining Uncontested Seats in Congress and State Legislatures," *American Politics Research* 49(3) (2021): 247–258, doi:10.1177/1532673X20960565 — on why parties decline to field candidates — one mechanism potentially contributing to Appendix E's contest gap, which (as Appendix E says) does not observe filing decisions and cannot single any mechanism out.
- **Political geography versus intentional gerrymandering.** Chen & Rodden, "Unintentional Gerrymandering: Political Geography and Electoral Bias in Legislatures," *Quarterly Journal of Political Science* 8(3) (2013): 239–269 — residential concentration produces seat/vote bias without deliberate line-drawing. Stephanopoulos & McGhee, "Partisan Gerrymandering and the Efficiency Gap," *University of Chicago Law Review* 82 (2015): 831–900 — the source of the efficiency-gap measure and of the benchmark Appendix E reports against.
- **Off-cycle timing and who is left deciding.** Anzia, *Timing and Turnout: How Off-Cycle Elections Favor Organized Groups* (University of Chicago Press, 2014).
- **Top-two and primary reform.** McGhee & Shor, "Has the Top Two Primary Elected More Moderates?" *Perspectives on Politics* 15(4) (2017): 1053–1066, doi:10.1017/S1537592717002158 — addresses candidate *moderation* under top-two, with mixed and jurisdiction-dependent findings; it is cited here for institutional context and does not speak directly to whether top-two restores general-election competition.
- **Safe-seat prevalence, advocacy literature.** FairVote's *Monopoly Politics* series reports projected U.S. House competitiveness and is the best-known non-academic statement of the prevalence claim this paper measures at state-chamber level. **It is deliberately not cited to a specific edition, and nothing here rests on it.** *(Resolved 2026-08-16 by removing the dependency rather than by supplying a pin.)* The series is issued in successive editions with differing projection methods and district universes, so citing one edition as "the" figure would import a number this paper cannot reproduce and does not need — every prevalence figure in this paper is derived from certified results, and the FairVote comparison would be to a *projection* of federal districts rather than an observed count of state ones. The placeholder that stood here, promising an edition and archive location before publication, was itself the defect: a citation owed is a citation that should either be paid or dropped, and this one is not load-bearing enough to pay.

Appendix E — Robustness, and exploratory cuts

Threshold sensitivity. The not-close share across margin cutoffs. The Washington rows come from `scripts/diag_seat_competition.py` on the corrected universe, using the same code path as the headline, so the ≥10 column reproduces Dimension 1 exactly; the other chambers come from `scripts/diag_safe_seat_robustness.py`.

scope	≥5pt	≥8pt	≥10pt	≥12pt	≥15pt	seats

scope	≥5pt	≥8pt	≥10pt	≥12pt	≥15pt	seats
WA all seats	92.5%	88.0%	**83.5%**	77.4%	73.7%	133
WA House	95.9%	91.8%	**87.8%**	80.6%	76.5%	98
NY Assembly	94.0%	92.0%	88.0%	85.3%	82.0%	150
TX House	98.0%	96.0%	94.0%	91.3%	86.0%	150
ID House	95.7%	92.9%	92.9%	91.4%	90.0%	70

Even at a stringent 15-point cut, **74–90%** of seats are not close; at a loose 5-point cut, 92–98%. Across every tested cutoff from 5 to 15 points, a large majority of seats remain not close.

Both New York rows — this sweep and the four-state table above — are now on the complete 150-seat chamber, and their ≥10-point cells agree at **88.0%**. `diag_safe_seat_robustness.py` reads the same pinned AD-23 file as the verifier rather than carrying its own copy, so the two code paths cannot drift to different denominators; each also refuses to run if the gap it is closing is ever not exactly one seat.

The contest gap. Comparing each chamber's actual not-close share with the share of its districts that are ≥10-point safe by 2024 presidential lean. The Washington figure uses the corrected House number:

	actual	presidential-lean safe	gap
WA House	87.8%	79.6%	**+8.2 pp**
TX House	94.0%	84.0%	**+10.0 pp**
ID House	92.9%	94.3%	−1.4 pp

In Washington and Texas, legislative contests are substantially less competitive than presidential lean alone would predict — about ten points less — while in deep-one-party Idaho the gap vanishes. (New York cannot be tested: its loaded presidential data predates the 2022 Assembly lines.)

What this comparison does not identify. An earlier version of this passage read the gap as candidate non-entry — "parties decline to field candidates where the presidential numbers say a seat is winnable" — and that is more than the statistic supports. The gap compares two aggregate shares; it does not observe who filed. A district can carry both major parties on the ballot and still finish twenty points apart for reasons that have nothing to do with entry: incumbency, candidate quality, spending, differential turnout, or simple office-specific voting by people who split their ticket. Non-entry is *one* mechanism consistent with the gap, and the uncontested-seat literature (Squire 2000; Burden & Snyder 2021) establishes that it operates somewhere — but it does not establish that it produces *this* +8.2 and +10.0. Decomposing the gap by whether the district was single-candidate, lacked one major party, or ran both parties to a wide margin would test it directly, on data already loaded. That is the sequel, and it is not attempted here.

Exploratory: seat/vote ratio and partisan symmetry. *This section is descriptive and should be read as a prompt for further work rather than a finding.* Measured against each state's 2024 presidential two-party vote, not-close seats over-represent the locally dominant party, and the gap widens with the state's lopsidedness: Washington **+2.1** points, Texas **+3.3**, Idaho **+18.9**. Two basis notes, because the same comparison has been stated two ways. These are **lower-chamber** figures, so Washington is its House (53 D / 33 R of 86 not-close seats = 61.6% D against a 59.5% Democratic presidential share); on the all-seats universe Appendix A quotes, the split is 68 D / 43 R = 61.3% D, a **+1.8** gap. Texas's holding party is the **observed** party from certified returns, not the retired presidential-lean imputation — that imputation gave 51 D / 90 R and a +6.9 gap, and was additionally circular, since it derived party *from* presidential lean and then compared it *against* presidential lean. That pattern is **consistent with** packing, and equally consistent with residential geography, district-boundary effects, turnout differences, and the definition of "safe" used here; single-member districts carry no proportionality expectation, and this comparison examines only safe seats rather than the whole distribution. It cannot distinguish those mechanisms or establish a counterfactual seat distribution. New York is omitted from the series: its only loaded presidential vote is 2020, against 2022 Assembly lines. Under the efficiency-gap specifications reported by `scripts/diag_efficiency_gap.py`, no state exceeds the ~8-point level Stephanopoulos & McGhee (2015) discuss (the largest is Texas at 6.5% on the wasted-vote form). Those values are computed from **district-level presidential two-party vote**, as a measure of asymmetry in the map, rather than from each chamber's own legislative returns as in the original specification; the script computes more than one formulation and the values differ (Texas is 6.5% wasted-vote against 0.1% simple), the metric is sensitive to turnout distribution, New

York's row again crosses a redistricting, and none of this determines whether a map is an outlier against neutral ensembles or was drawn with intent. Establishing that requires map-simulation work this paper does not attempt.

Exploratory: comparison with the project's forecast. This project's forecast bands 53 of 59 Washington districts (90%) as ≥10-point safe for 2026, against 83.5% of seats measured as not-close on 2024 results. These are **different units, different years, and different denominators** — districts versus seats — so the similarity is a loose aggregate consistency check, **not a validation**. A real validation would predict historical elections without using their outcomes and report classification accuracy, calibration, and false-safe rates on the same unit.

Appendix F — The Texas backfill

Texas is the one state where this paper adds rows rather than only counting them.

The problem. The Texas Legislative Council's canvass-grade VTD returns omit uncontested races: when a seat is unopposed no precinct tally is published. The warehouse therefore carried **96 of 150** TX House districts, and the missing 54 were missing *because* they were uncontested — exactly the category this paper counts. Left alone, the omission biases Texas toward looking more competitive than it is.

The original construction, now retired. The 54 absent districts were backfilled as *no major-party choice*, with holding party assigned from each district's 2024 presidential lean using the on-disk TLC r206 report (`planh2316_r206_election24g.xls`). Script: `scripts/diag_tx_safe_seat_backfill.py` . The results database was not mutated. The uncontested classification from that step survives verification below; the imputed party does not, and no figure in this paper now depends on it.

Verified against a certified source. The backfill was checked seat by seat against the Texas Secretary of State's election-night results service, which publishes every race including uncontested ones with candidate name, party and votes (`scripts/build_tx_house_candidates.py` → `data/raw/tx/2024_tx_house_candidates.csv` ; audit in `scripts/diag_tx_backfill_verification.py` → `reports/tx_backfill_verification.csv`).

The match is exact. The certified source shows **54 single-candidate districts**, and they are precisely the 54 the TLC returns omit — no district was backfilled that was actually contested, and no uncontested district was missed. The backfill is confirmed in full, not merely for a named subset. As an external check on the extraction, the certified data also puts the 2024 chamber at 62 D / 88 R, matching the seated Texas House.

Why the TLC data could never have settled this. It omits uncontested races at *every* stage, not just the general: all 14 districts on the press-reported unopposed list appear in **neither** 2024 party primary, because those candidates were unopposed there too. Absence is a property of the whole dataset, so no amount of work inside it could distinguish "uncontested" from "missing".

Holding party is now observed, and the old imputation was wrong. Party for the 54 comes from the certified returns rather than presidential lean. Retrospectively, the retired imputation was **wrong in 5 of 54** — HD 35, 36, 40, 42 and 144, every one a district Trump carried while a Democrat held the seat, mostly Rio Grande Valley. The observed split of the backfilled seats is **36 D / 18 R**, against 31 D / 23 R imputed. Correcting it moves the chamber's not-close split from the previously reported 51 D / 90 R to **56 D / 85 R**.

The headline was unaffected throughout. Not-close remains **141 of 150 = 94.0%** on certified data, identical to the backfilled figure, because that count needs only the seats to have been uncontested — which is exactly what the certified source confirms. Without the backfill, on the 96 contested-skewed districts, Texas would read 90.6% and just 7.3% without a D-vs-R option, visibly biased toward competitiveness.

Appendix G — What changed in this revision, and why

An adversarial review of the prior version identified three defects. All were confirmed against the code and data, and all are corrected here.

1. The seat universe was incomplete. The prior analysis derived races from the precinct-results table. King County is essentially absent from Washington's statewide *precinct* files for 2016 and 2018 — about 190 of 330,845 rows in 2016 — so every King legislative race silently vanished: 24 House seats per year, districts 5, 11, 33, 34, 36, 37, 41, 43, 45, 46, 47 and 48. The warehouse held 74 of 98 House seats in both years. A separate format variant (2020 LD15 spelled "Representative, Position N" rather than "State Representative Pos. N") dropped two more. The certified summary files carry every race and were already on disk, so the fix required no new data — only building the universe from a certified race list and asserting it against the statutory chamber size.

2. "Same-party" was misclassified. The rule was `same_party if (d == 0 or r == 0)` , which captures *any* race lacking a D-vs-R pairing — including D-vs-independent and R-vs-Libertarian. Of the 23 races the prior version labelled same-party in

2024, **seven were not same-party at all.** Relatedly, winner party was inferred by comparing aggregate D and R totals, so a race with neither would have been scored Democratic; no such race occurred in 2024, but the rule was wrong and is now taken from the leading candidate.

3. Two questions were collapsed into one. Every same-party general was scored non-competitive regardless of margin. That is a statement about partisan availability presented as a statement about competitiveness.

Effect on the headline figures:

year	prior "non-competitive"	corrected "not close"	change
2016	90.7%	88.1%	−2.6
2018	**75.0%**	**78.9%**	**+3.9**
2020	84.1%	83.6%	−0.5
2022	87.1%	86.5%	−0.6
2024	**85.0%**	**83.5%**	**−1.5**

The most consequential change is 2018. The prior version reported a dramatic dip to 75% in the blue-wave year; on the complete universe the dip is to **78.9%** — still the least lopsided cycle in the series, but a good part of the apparent drop was 24 missing King County seats, which are disproportionately safe and Democratic. The headline moves from "roughly 85%" to **"roughly 84%"**, and the decade range tightens from 75–91% to 79–88%.

The 2024 no-major-choice count moves from 46 to **47**, and is now correctly described: 23 single-candidate, 15 same-party, 9 major-versus-minor. *(Correct as of the 2026-07-27 revision and superseded by the 2026-08-08 party-string audit, which returns the count to 46 on a different decomposition — 23 single-candidate, 16 same-party, 7 major-versus-minor. The coincidence of arriving back at 46 is worth naming: it is not a reversal, and the two 46s do not describe the same set of seats.)*

Also withdrawn: the claim that seats were "decided before November," which observed November margins cannot establish; and the assertion that the safe-seat ratio is a "packing signature," now reported as a descriptive comparison consistent with several mechanisms.

The second pass, 2026-07-28

The rebuild above fixed the argument. A later pass audited every printed figure against the script that claims to produce it — a different exercise, and it found five numbers that no longer matched their own derivation, plus one table that had none.

1. A misspelled party string made a real two-party race read as no-choice. The 2020 certified file records Legislative District 8's Position 1 Democrat as preferring the "Democractic" party. The strict rule matches "Democratic" and "Democrat", so the string fell through to *other* and a Klippert-versus-Regev general — 51,981 to 26,979, a Republican against a Democrat — was classified Republican-versus-other. The 2020 no-D-v-R share moves **27.6% → 26.9%**, D-v-R from 97 to 98, R-v-other from 5 to 4, and the loose-rule delta from 2.2 to 1.5. Dimension 1 is untouched: the seat was a 31.7-point blowout either way. The fix is applied in both specifications, because a transcription error is not the kind of thing a strict-versus-loose sensitivity test is meant to be sensitive to.

2. The primary/general table had no derivation. The five-cycle series was reported as "recomputed on the corrected universe," but the recomputation lived in no script: `diag_seat_competition.py` had no primary logic at all, and the only script that computed the ratio was the superseded pre-rebuild one, which covered two cycles on the old universe. The computation is now part of `diag_seat_competition.py`, which also reports how many seats matched and fails the run if any did not. Four of the five cycles reproduce the published figures exactly; 2020 comes out **61.5%** rather than 61.6%.

3. Appendix E carried two stale exploratory gaps. Washington's +2.6 came from the retired 88.8% universe; on the certified one it is **+2.1** (House basis) or **+1.8** (all seats), and the paper now states which basis each figure uses, because Appendix A quotes the all-seats split against the same presidential share. Texas's +6.9 was computed on the 51 D / 90 R imputation that Appendix F retires as wrong in 5 of 54 seats; on observed party it is **+3.3**. The ordering the section describes — the gap widening with a state's lopsidedness — survives both corrections.

4. New York's ≥12-point cell was mistranscribed, 85.2% for 85.9% (127 seats for 128). *(Both figures are on the 149-seat chamber this pass ran against. Superseded by the 2026-08-08 supplement of Assembly District 23: the count stays 128 — AD-23 was decided by 0.046 points and is close at every threshold — so the denominator moves 149 → 150 and the cell reads 85.3% in Appendix E. The same denominator change moved every other New York threshold cell, which the revision note below now records.)*

5. The comparison states are scored by a shortcut that had not been checked. Their code treats any seat lacking a D-vs-R

option as not close regardless of margin — the conflation defect #3 removes for Washington. Re-scoring all 129 such seats on their top-two margin finds **none under ten points**, so every comparison-state percentage is identical under either rule. The check now runs as part of `diag_safe_seat_robustness.py` rather than being assumed, and the four-state section states the result.

Nothing in the abstract changed. The 2016–2024 not-close series, the 2024 headline of 111 of 133 seats, the 47-seat no-choice count, the cross-tab, and Appendix F's Texas verification all reproduce unchanged. *(Historical as of the 2026-08-08 audit: the no-choice count is now 46 and the cross-tab moved with it. The 111-of-133 headline is genuinely unchanged — margins do not depend on party.)*

> **Revision note (2026-08-08) — third revision, the party-string audit.** Party classification was a regex, and a regex cannot report what it fails to match. Enumerating every distinct party string in the five certified files — 32 of them — found **six races** misclassified across four cycles, five of them never suspected: `G.O.P.` (2018), `G.O.P` without the trailing period (2016), `R` twice (2020), `MAGA Republican` (2024) and `Culture Republican` (2024). Each read as a minor party, so a genuine major-party contest was classified as major-versus-other.
>
> **Figures that moved:** 2018 no-D-v-R 27.1% → **26.3%**; 2020 26.9% → **25.4%**; 2024 35.3% → **34.6%** (47 seats → **46**, decomposing 23 / 16 / 7 rather than 23 / 15 / 9); the five-cycle range 27–49% → **25–49%**; same-party generals 15 → **16**; the 2024 cross-tab. **Dimension 1 did not move at all** — margins are computed between candidates regardless of party — so 111 of 133 (83.5%) stands, as does the whole not-close series.
>
> **What replaced the regex:** both specifications are now enumerated string by string in `docs/reference/wa_party_strings_2016-2024.csv`, each row carrying its reason, and an unlisted string raises rather than defaulting to "other". A third specification was added at the same time (see §Sensitivity), separating orthographic normalisation from the party-family grouping, because folding "Culture Republican" in with "Republican" is a researcher's category and not the state's.
>
> Also in this revision: New York Assembly District 23 supplied from the certified NYSBOE contest, retiring the 149-seat bound; Appendix E's contest gap no longer read as candidate non-entry; "foreclosure" removed from the descriptive results; and the Senate universe claim narrowed to what is independently asserted.
>
> **What supplying AD-23 moved (added 2026-08-10, round 6; sourcing corrected round 7).** The seat is close at every threshold — 0.046 points — so the not-close *counts* are unchanged and only the denominator moves, **149 → 150**. Every one of Appendix E's five New York threshold cells rests on that denominator, not just the ≥10 one this note originally named. Stated as the counts, which are what the source actually fixes:
>
> | cut | not-close count | on 149 | on 150 (published) |
> |---|---|---|---|
> | ≥5 | 141 | 94.6% | **94.0%** |
> | ≥8 | 138 | 92.6% | **92.0%** |
> | ≥10 | 132 | 88.6% | **88.0%** |
> | ≥12 | 128 | 85.9% | **85.3%** |
> | ≥15 | 123 | 82.6% | **82.0%** |
>
> **Provenance, because two of these are recorded and three are reconstructed.** The ≥10 (88.6%) and ≥12 (85.9%) figures are corroborated in this paper's own text — the limits section and second-pass item 4. The ≥5, ≥8 and ≥15 values on 149 appear in **no** record; they are computed here from the published 150-based percentages, each of which determines its count uniquely (94.0% ⇒ 141, since 140/150 = 93.3% and 142/150 = 94.7%). So the *counts* are sound and the 149-based percentages are arithmetic rather than history: whether the 2026-07-28 pass printed them at all is not something this record can establish.

End note — data, reproduction, and series

```
# The rebuilt classification — universe assertion, both dimensions, sensitivity,
# and the primary/general medians. Every Washington figure in the paper comes from here.
python scripts/diag_seat_competition.py        # writes reports/seat_competition.csv

# Four-state lower-chamber count and the Texas completion:
python scripts/diag_safe_seat_states.py        # NB: its WA cell (88.8%) and TX safe split
                                               # (51/90) are SUPERSEDED and labelled so in
                                               # its output; see Appendix C
python scripts/diag_tx_safe_seat_backfill.py       # Appendix F backfill
python scripts/build_tx_house_candidates.py        # certified TX candidate list (all 150)
python scripts/diag_tx_backfill_verification.py    # Appendix F seat-by-seat verification
                                                   # -> reports/tx_backfill_verification.csv

# Robustness and the exploratory cuts:
python scripts/diag_safe_seat_robustness.py    # Appendix E thresholds + contest gap
python scripts/diag_safe_seat_party_ratio.py   # seat/vote ratio
python scripts/diag_efficiency_gap.py          # partisan-symmetry diagnostic
```

All inputs are published election returns. See `data-sources-and-reproducibility.md` for the source ledger and `electoral-health-audit-log.md` for the verification ledger of expected values.

Companion to the electoral-health series lead, `who-decides-washington.md`, with `who-decides-new-york.md` and `who-decides-idaho.md` (party-resolved electorates), `donor-class-and-the-electorate.md` (who funds them), `cross-state-fec-money.md` (the four-state money layer), and `does-money-move-votes.md` (whether that money moves margins).